# ECG Interpreter

THE CHALLENGE

# Nigeria's Cardiac Care Gap

**1:10K**
Cardiologist-to-patient ratio[3]

**<300**
Echo machines nationwide[2]

**60%**
CVD deaths in LMICs — preventable[1]

**216M**
Nigeria's population[1]

## The Diagnostic Gap

- Most patients diagnosed **late** or not at all
- Specialists concentrated in Lagos & Abuja
- Referral chains are long and costly[2]

## The Opportunity

- ECG machines already in rural PHCs (₦480,000–₦1.6M)[8]
- ECG costs **₦4,800–₦12,800** vs ₦320,000–₦640,000 for echo[8]
- AI bridges the **skills gap** without new hardware

THE SOLUTION

# ECG Interpreter — What It Does

## Paper ECG Friendly

- Accepts **smartphone photos** of printed ECGs
- AI digitiser converts image → waveform
- Works with any 12-lead machine

## Fully Offline

- Runs on a **laptop or tablet**
- No internet or cloud subscription needed
- ~121 MB total model weight

## Clinical Decision Support

- Flags **HIGH / MOD / LOW** risk in <30 s
- Evidence-weighted differential diagnoses
- Augments — does not replace — the clinician

**Real scenario:** PHC nurse in Kano photographs ECG strip → AI flags *Reduced EF + Pulmonary HTN* (HIGH risk) → Urgent referral issued. Cost: **₦4,800 ECG + a smartphone photo.**[8]

ARCHITECTURE

# End-to-End Pipeline

**1. Input**
JPEG / PNG / PDF

→

**2. Digitiser**
Image → Waveform

→

**3. Features**
NeuroKit2[7]

→

**4. Dual AI**
Two Neural Nets

→

**5. Report**
Risk + Diagnoses

## Model 1 — PTB-XL ECG Classifier[4]

Input: 12-lead ECG · 12 × 2,500 samples @ 250 Hz

↓

1D ResNet Backbone (ECGEncoder1D) — 4 stages, ~5M params

↓

256-dim ECG Embedding (Global Avg Pool)

## Model 2 — EchoNext Structural Detector[6]

ECG Branch → 256-dim | Tabular (7 features) → 128-dim

↓ concatenate → 384-dim

Fusion MLP: 384 → 256 → 64

↓

12 Structural Conditions (sigmoid output)

DATA

# Training Datasets

## PTB-XL v1.0.3[4]

| Attribute | Value |
|---|---|
| Source | PhysioNet / Univ. Basel |
| Total records | **21,799** |
| Train / Val / Test | 17,439 / 2,180 / 2,180 |
| Sampling rate | 100 Hz → resampled 250 Hz |
| Labels | SCP codes — dual cardiologist verified |
| Size | 601 MB |

12 binary classes: MI territories · BBB · AF · LVH · ST-T changes · Normal

## EchoNext (CUIMC)[6]

| Attribute | Value |
|---|---|
| Source | Columbia University Medical Centre |
| Ground truth | Echocardiography findings |
| ECG shape | (N, 1, 2500, 12) @ 250 Hz |
| Tabular features | Sex · HR · PR · QRS · QTc · Age |
| Labels | 12 binary structural conditions |
| Best checkpoint | 61 MB |

Multi-modal fusion: ECG waveform + 7 clinical

features → +3–5% AUROC gain

TRAINING

# Training Configuration

## Shared Hyperparameters

| Parameter | Value |
|---|---|
| Optimizer | AdamW |
| Learning rate | $3 \times 10^{-4}$ · cosine annealing |
| Weight decay | $1 \times 10^{-4}$ |
| Loss function | BCEWithLogitsLoss (class-weighted) |
| Hardware | Apple M-series · MPS GPU |

**~8 h**
EchoNext (15 epochs)

**~3 h**
PTB-XL (17 epochs)

## Model-Specific Settings

| | PTB-XL | EchoNext |
|---|---|---|
| Batch size | 64 | 32 |
| Max epochs | 50 | 30 |
| Early stopping | patience 10 | patience 7 |
| Best epoch | **7** | **7** |
| Dropout | — | 0.2 |

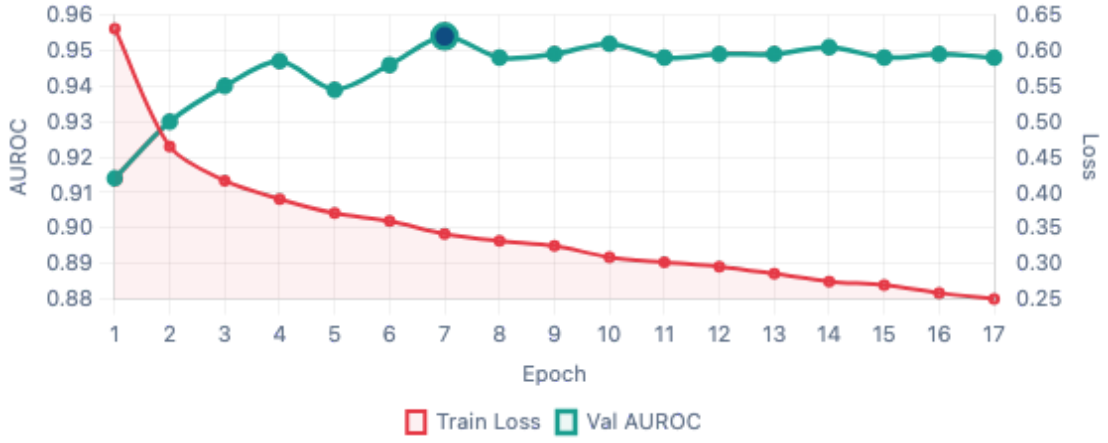
## Data Augmentation (EchoNext)

- **Gaussian noise** — $\sigma \approx 1\%$ of signal amplitude
- **Amplitude scaling** — $\times 0.9$ to $\times 1.1$
- **Baseline wander** — low-frequency drift simulation
- **Class weighting** — up-weights rare conditions

RESULTS

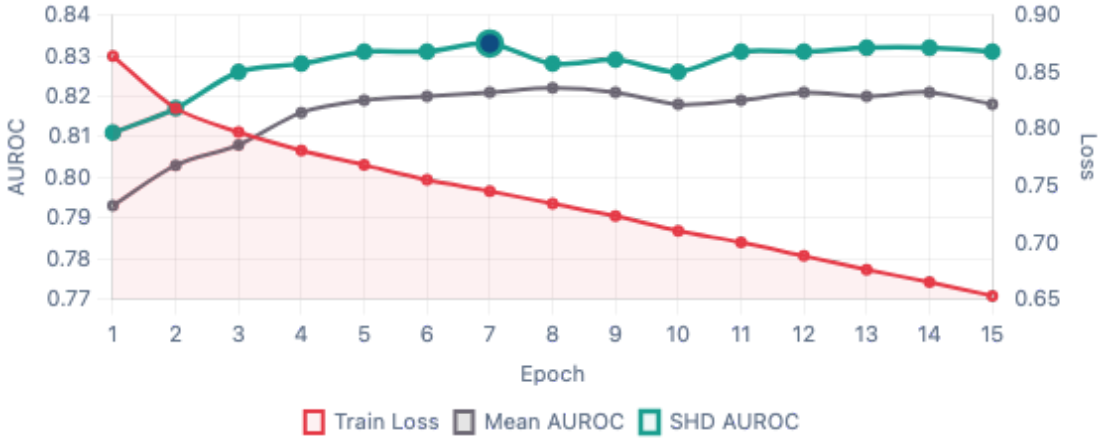
# Training Curves — Loss & Validation AUROC

## PTB-XL Classifier




| 0.914 | 0.954 | 0.948 |
|---|---|---|
| Epoch 1 | Best — Ep 7 ★ | Epoch 17 |

## EchoNext Structural Detector




| 0.793 | 0.833 | 0.821 |
|---|---|---|
| Epoch 1 | SHD — Ep 7 ★ | Mean Ep 7 |

Best checkpoints saved at Epoch 7 for both models — cosine LR annealing + early stopping prevented overfitting

PERFORMANCE

# Validation AUROC — All 24 Conditions

## PTB-XL — 12 ECG Diagnoses[5]

## EchoNext — 12 Structural Conditions[6]

| | | |
|---|---|---|
| ★ Any SHD (composite) 0.833 | Reduced EF (LVEF≤45%) 0.815 | LV Wall Thickening 0.807 |
| Aortic Stenosis 0.811 | Mitral Regurgitation 0.812 | Pulmonary HTN 0.821 |
| RV Dysfunction 0.816 | Pericardial Effusion 0.819 | TR Velocity 0.815 |
| Aortic Regurgitation | Tricuspid Regurg. | Pulm. Regurgitation |

LIVE DEMO

# Interface Walkthrough

## Input

Upload ECG image (JPEG / PNG / PDF)

## Digitised Waveform

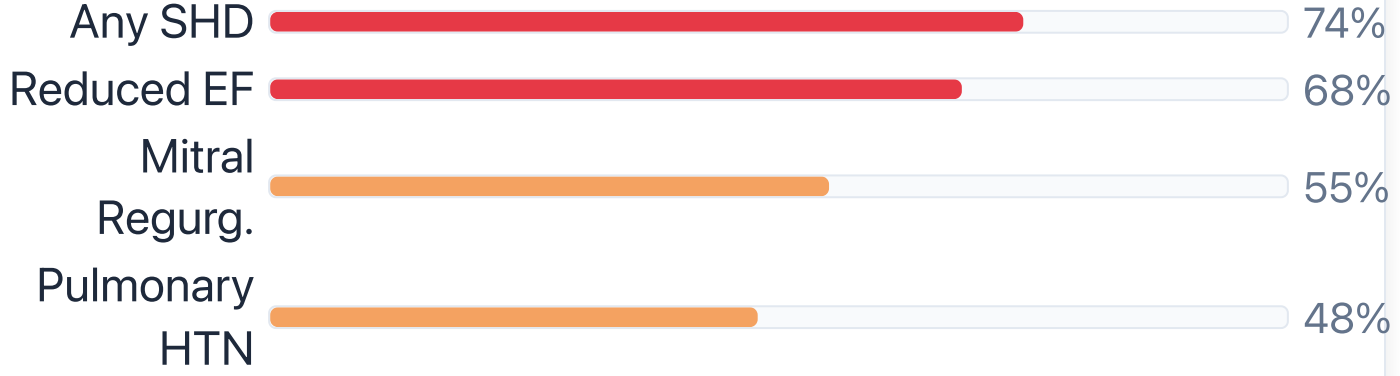
## Extracted Features

| | |
|---|---|
| Heart Rate | **87 bpm** |
| QRS Duration | **124 ms ↑** |
| PR Interval | 168 ms |
| QTc Interval | 445 ms |

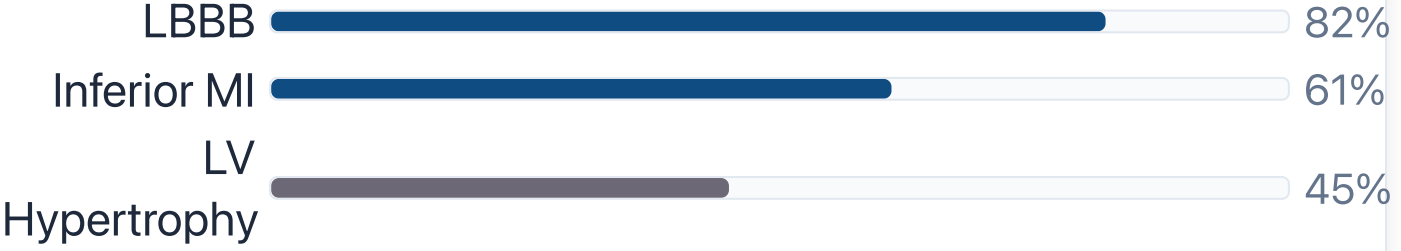
## Risk Stratification

**▲ HIGH RISK — SHD Probability: 74%**

### Structural (EchoNext)




### ECG Diagnoses (PTB-XL)



| | |
|---|---|
| Patient | 58 yo / Male |

⚠ **Urgent Referral:** Dilated cardiomyopathy + LBBB pattern. Echo confirmation advised.

Launch locally: `streamlit run app/app.py`

TECHNOLOGY

# Tools & Libraries

### PyTorch 2.0+
Deep learning · Model training & inference · MPS / CUDA / CPU support

### Streamlit
Web application UI · File upload · Real-time results display

### Plotly
Interactive waveform & probability charts in the app

### Pandas / NumPy
Dataset loading · Feature engineering · Training log analysis

### Scikit-learn
AUROC computation · ROC analysis · Evaluation metrics

### NeuroKit2[7]
ECG feature extraction · R-peak detection · QRS / PR / QTc intervals

### OpenCV + Pillow
Image preprocessing · Deskewing · Grid removal · Trace isolation

### SciPy
Signal resampling (100→250 Hz) · Filtering · Baseline correction

### WFDB
PhysioNet data loading · PTB-XL record parsing · ECG format I/O

### pdf2image / pypdf
PDF ECG import · Page-to-image conversion

### Matplotlib
Training curve plots · Analysis & validation graphs

### tqdm
Training progress bars · Batch & epoch iteration monitoring

CLINICAL AI

# Diagnosis Engine — How the AI Reasons

## 3-Source Evidence Fusion

PTB-XL Model Probabilities (ECG pattern recognition)

+

Morphological Features (ST segments, Q waves, rhythm)

+

EchoNext Structural Probabilities

↓

Evidence-Weighted Scoring Engine

↓

## Evidence Logic

- **Anchors** — required findings to consider a diagnosis
- **Supporters** — boost confidence by +10–25%
- **Against** — dampen score (e.g. narrow QRS → not LBBB)

```
confidence = (anchor + support_bonus) × dampen_factor
```

## Risk Thresholds

Ranked Differential Diagnoses + Confidence

| ≥60% | 35–59% | <35% |
| --- | --- | --- |
| HIGH — urgent | MODERATE | LOW risk |

IMPACT

# Deployment Strategy for Nigeria

## Primary Health Centres

- Nurse photographs ECG → instant AI triage
- No on-site specialist required
- Flags urgent referrals automatically
- 35M+ PHC patients/year[2]

## District Hospitals

- Specialist-level AI for general practitioners
- Reduces unnecessary referrals
- Tele-cardiology ready output
- Bridges rural–urban gap

## Diagnostic Centres

- 50–200 ECGs/day processed[2]
- Interpretation: 15 min → **30 seconds**
- Standardised, documentable reports
- No part-time cardiologist needed

**97%**
Cost reduction vs specialist

**30s**
Per-ECG AI interpretation

**24/7**
Offline availability

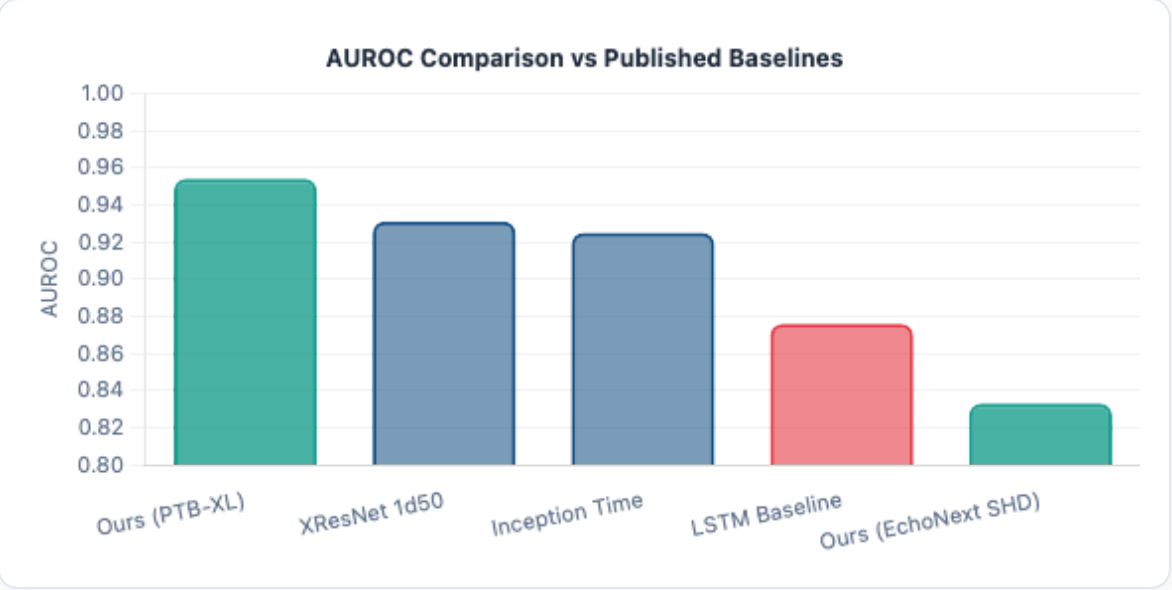
**₦0**
Per-analysis after setup[8]

BENCHMARKING

# Performance vs Published Baselines[5]

## PTB-XL Comparison[5]

| Model | AUROC |
|---|---|
| **Ours (1D-ResNet)** | 0.954 |
| XResNet1d50 | 0.931 |
| InceptionTime | 0.925 |
| LSTM Baseline | 0.876 |




Competitive AUROC achieved in **<12 total hours** on a consumer laptop — no cloud GPU required

Echo-image CNNs reach ~0.89 AUROC but require expensive hardware[6] — our

LIMITATIONS & ROADMAP

# Honest Assessment & Next Steps

## Population Gap

Trained on European & US data. Nigerian-specific morphology (RHD, African cardiomyopathy) not yet represented[2]

→ Fine-tuning study planned with LUTH / UCH dataset

## Image Quality

Digitiser requires clear, landscape ECG photos (min 800×600 px)

→ In-app quality guidance before inference

## Clinical Reminder

## Future Roadmap

### Near-term

- Nigerian dataset fine-tuning
- Hausa / Yoruba / Igbo UI
- Model quantisation for low-spec devices

### Mid-term

- HL7/FHIR EMR integration
- Multi-centre Nigeria validation study
- NAFDAC regulatory submission

AI output is **decision support only** — licensed clinician must confirm before treatment

## Long-term

- Pan-Africa ECG consortium
- Transformer (ECG-BERT) upgrade
- Federated learning across hospitals

```
streamlit run
app/app.py
```

REFERENCES

# Index of Sources


[1] World Health Organization. *Global Health Estimates: Leading Causes of Death*. WHO; 2020. Available: who.int/data/gho

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[3] Adeyemi AA, et al. "Cardiology workforce in Nigeria: a situational analysis." *Pan African Medical Journal*. 2019;34:192.

[4] Wagner P, Strodthoff N, Bousseljot R-D, et al. "PTB-XL, a large publicly available electrocardiography dataset." *Scientific Data*. 2020;7(1):154. doi:10.1038/s41597-020-0495-6.

[5] Strodthoff N, Wagner P, Schaeffter T, Samek W. "Deep Learning for ECG Analysis: Benchmarks and Insights from PTB-XL." *IEEE Journal of Biomedical and Health Informatics*. 2021;25(5):1519–1528.

[6] Ouyang D, He B, Ghassemi M, et al. "Video-based AI for beat-to-beat assessment of cardiac function." *Nature*. 2020;580(7802):252–256. (EchoNet/EchoNext architecture basis)

[7] Makowski D, Pham T, Lau ZJ, et al. "NeuroKit2: A Python Toolbox for Neurophysiological Signal Processing." *Behavior Research Methods*. 2021;53(4):1689–1696. doi:10.3758/s13428-020-01516-y.

[8] Central Bank of Nigeria. *Official Exchange Rate*. CBN; March 2026. Rate applied: **₦1,600 / USD**. Available: cbn.gov.ng. [USD costs converted: ECG $3–8 = ₦4,800–₦12,800; Echo $200–400 = ₦320,000–₦640,000; ECG machine $300–$1,000 = ₦480,000–₦1.6M]